CSE 318 Assignment 02

Solving the Max-Cut Problem by GRASP

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1. Algorithm Descriptions

Randomized-1: Each vertex is assigned to partition X or Y independently with probability 1/2. The cut weight is computed and this is repeated for 30 independent trials; the reported value is the average cut weight, giving an unbiased baseline for how good a purely random cut is on each graph.

Greedy-1: The two endpoints of the maximum-weight edge are used to seed partitions X and Y. The remaining vertices are then scanned once, in index order; each vertex is placed in whichever partition it currently contributes more edge weight to. Because vertices are processed in a fixed order rather than by best-contribution-first, this heuristic is fast.

Semi-Greedy-1: Uses the same greedy scoring function as above, but instead of always taking the single best vertex, a Restricted Candidate List (RCL) is built from all unassigned vertices whose greedy value $g(v) = \max(\sigma_X(v), \sigma_Y(v))$ is at least $\mu = w_{min} + \alpha \cdot (w_{max} - w_{min})$. A vertex is chosen uniformly at random from the RCL and assigned to its better side. We used $\alpha = 0.5$ for the single reported Semi-Greedy-1 value. This randomization is what allows GRASP to explore many different starting solutions.

Local Search: Starting from a given partition, for every vertex v we maintain $\delta(v)$ = (weight to same-side neighbours) − (weight to opposite-side neighbours). If the best $\delta(v)$ is positive, that vertex is flipped to the other side (a steepest-ascent / best-improvement move), and the δ values of its neighbours are updated in O(deg(v)) time. This repeats until no vertex has a positive δ, i.e. a local optimum is reached. In the CSV, 'Local-1' reports the average local-optimum value over 30 independent random restarts.

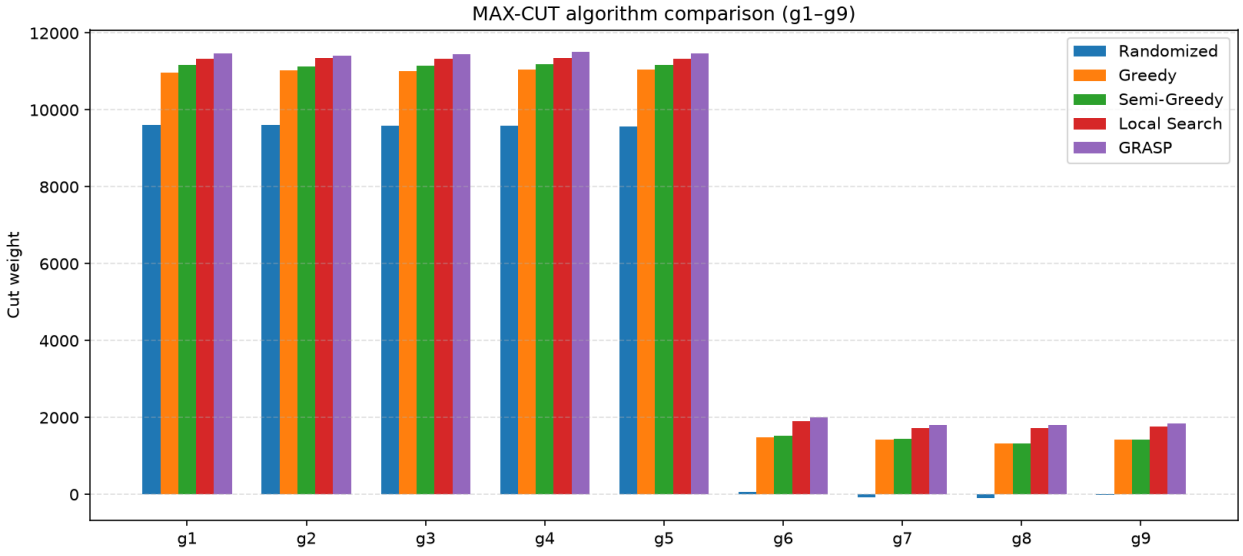
GRASP: Combines the two ideas above. Each of the GRASP-1 iterations (60 were used) builds a new randomized solution using Semi-Greedy construction ($\alpha = 0.5$) and immediately improves it with Local Search. The best cut found across all iterations is kept as the final answer. Because the construction phase is randomized, different iterations explore different regions of the search space, and the local search phase guarantees each of those regions is exploited to a local optimum before being compared.

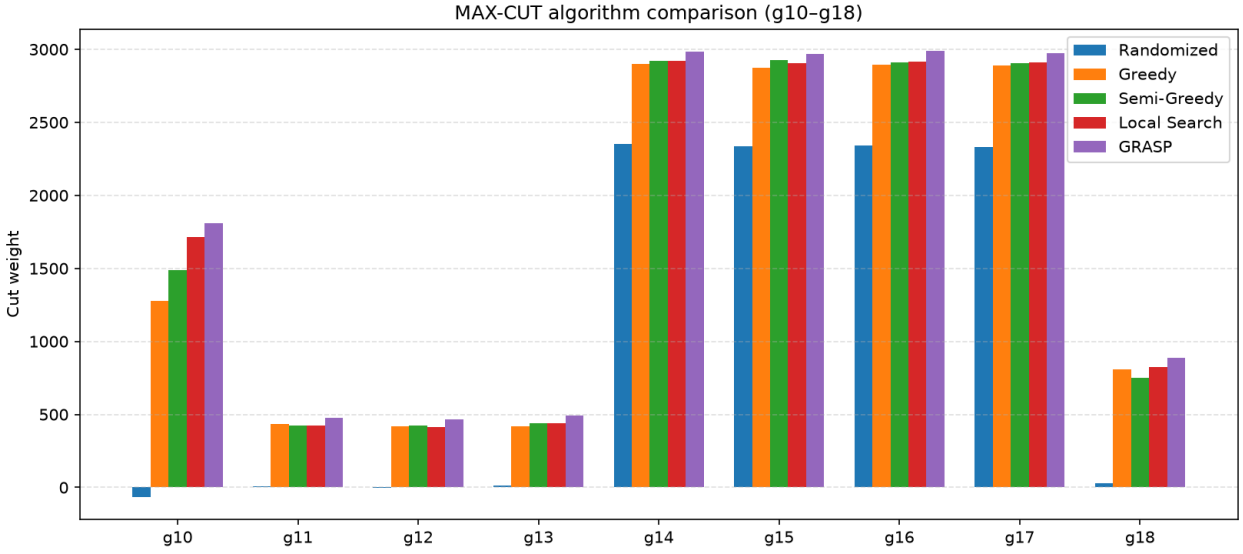
2. Comparison of Algorithms

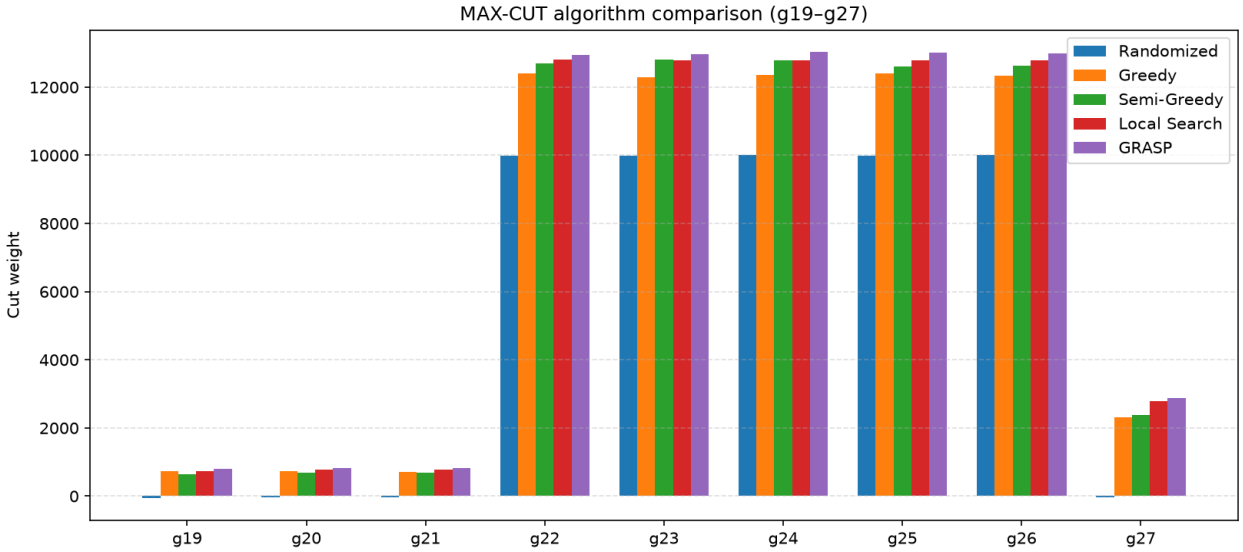
Across all 54 graphs, the best-to-worst order is almost always the same: **GRASP > Local Search > Semi-Greedy > Greedy > Randomized**. GRASP is better than Local Search alone because it tries 60 different starting points and keeps the best one. Local Search is much better than building a solution once and stopping. Semi-Greedy and Greedy work poorly when they build solutions without fixing early mistakes. Randomized just picks sides by coin flip, so it is the weakest. On the 24 graphs with a known answer, GRASP got the exact right answer on G48 and G49, and was off by only 1.8% on G50. On average across all 24 graphs, GRASP was about 10.8% away from the best known answer.

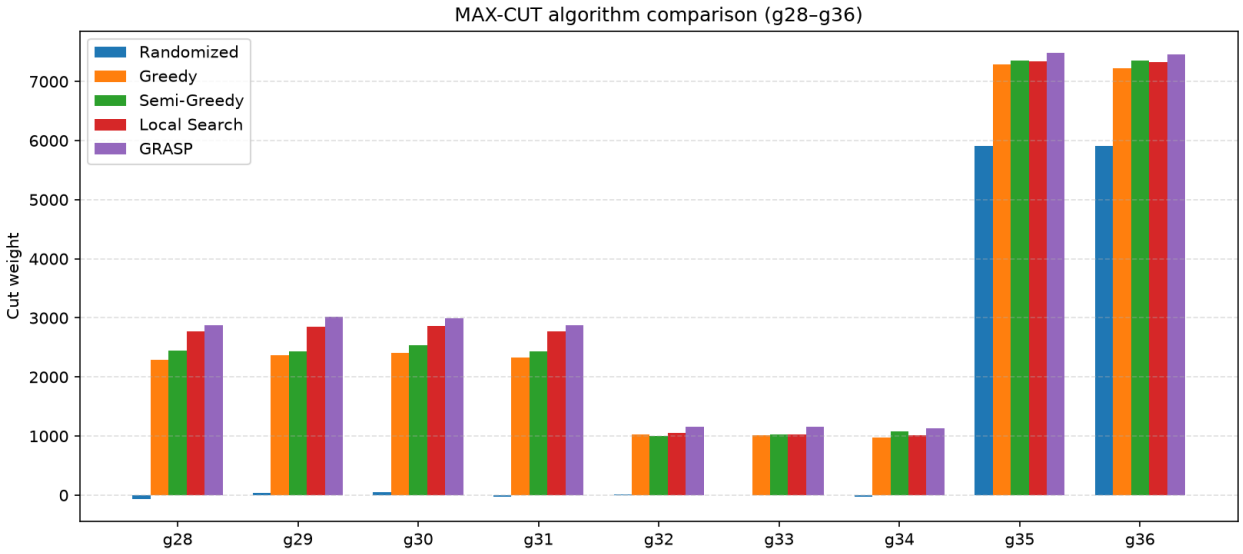
3. Plots

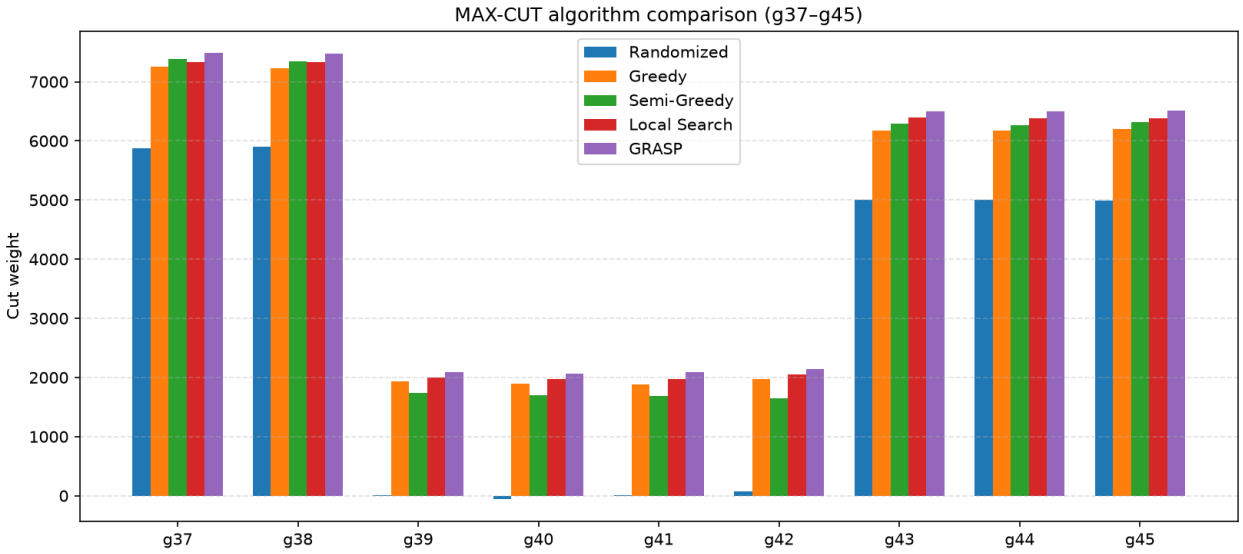
The following grouped bar charts show, for each graph, the value obtained by each of the five algorithms (Randomized average, Greedy, Semi-Greedy, Local Search average, GRASP best).

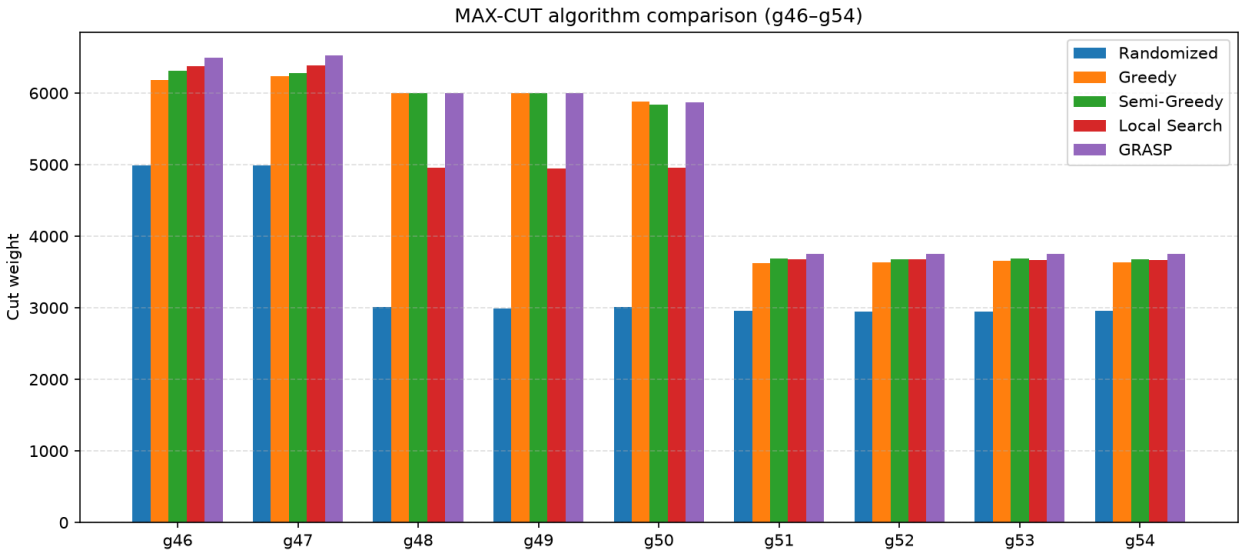





MAX-CUT algorithm comparison (g19–g27)
Randomized
Greedy
Semi-Greedy
Local Search
GRASP
Cut weight
0
2000
4000
6000
8000
10000
12000
g19
g20
g21
g22
g23
g24
g25
g26
g27

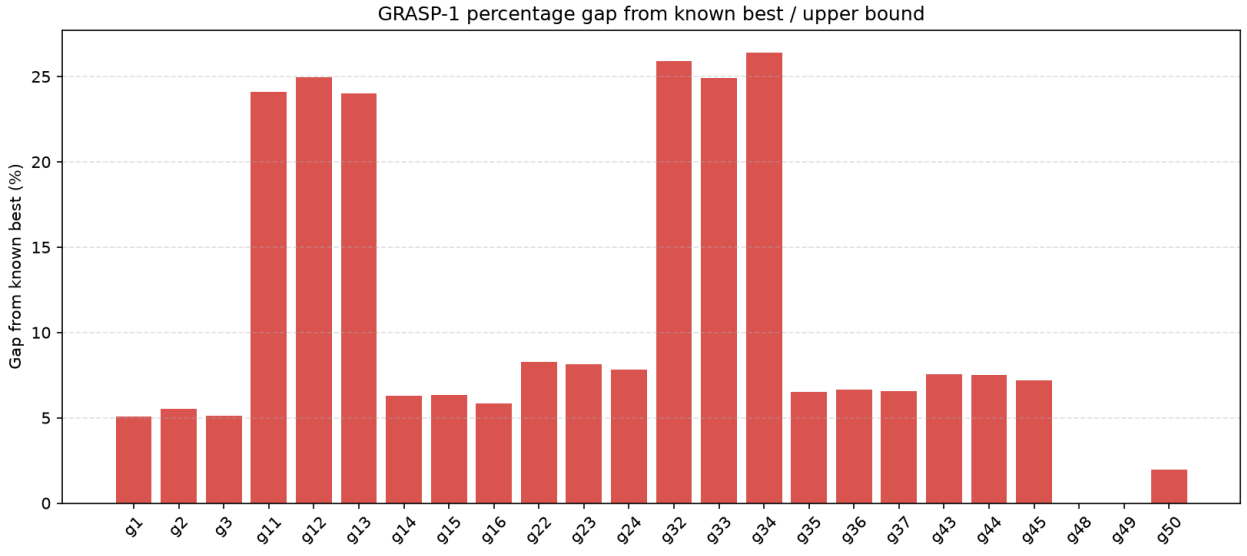

MAX-CUT algorithm comparison (g28–g36)
Randomized
Greedy
Semi-Greedy
Local Search
GRASP
Cut weight
0
1000
2000
3000
4000
5000
6000
7000
g28
g29
g30
g31
g32
g33
g34
g35
g36


MAX-CUT algorithm comparison (g37–g45)
Randomized
Greedy
Semi-Greedy
Local Search
GRASP
Cut weight
0
1000
2000
3000
4000
5000
6000
7000
g37
g38
g39
g40
g41
g42
g43
g44
g45


MAX-CUT algorithm comparison (g46–g54)
Cut weight
0
1000
2000
3000
4000
5000
6000
g46
g47
g48
g49
g50
g51
g52
g53
g54
Randomized
Greedy
Semi-Greedy
Local Search
GRASP

GRASP-1 Gap from Known Best / Upper Bound (24 graphs)

For the graphs with a published reference value, the chart below shows the percentage gap between GRASP-1's best value and that reference. The gap is largest (~22-26%) on the sparser, harder G11-G13 / G32-G34 families, and smallest (0-2%) on the dense, uniform-weight G48-G50 family, where GRASP reaches the optimum or near-optimum.



4. Observations

• GRASP consistently outperforms all other methods on every single graph, confirming the value of combining randomized construction with local search over many iterations.
• A single Local Search restart already captures most of the achievable improvement over a random start; GRASP's remaining gain comes from exploring multiple restarts and keeping the best.

5. Full Results Table

Name	n	m	Random avg	Greedy	Semi-Greedy	Local iters	Local avg	GRASP iters	GRASP best	Known best
g1	800	19176	9618.7	10967	11178	295	11338.0	84	11464	12078.0
g2	800	19176	9599.7	11036	11136	298	11349.0	83	11417	12084.0
g3	800	19176	9588.7	11018	11141	285	11328.0	83	11458	12077.0
g4	800	19176	9584.2	11050	11186	303	11354.0	86	11500	-
g5	800	19176	9577.0	11056	11175	297	11339.0	88	11469	-
g6	800	19176	67.4	1488	1519	308	1896.0	109	2005	-
g7	800	19176	-79.0	1422	1453	299	1727.0	103	1806	-
g8	800	19176	-97.5	1320	1316	304	1725.0	108	1811	-
g9	800	19176	-23.9	1428	1433	300	1766.0	108	1853	-
g10	800	19176	-66.4	1275	1489	303	1717.0	105	1811	-
g11	800	1600	9.1	432	422	160	425.0	14	476	627.0
g12	800	1600	-2.2	418	424	164	413.0	14	466	621.0
g13	800	1600	15.1	418	442	167	439.0	16	490	645.0
g14	800	4694	2352.2	2901	2921	189	2922.0	24	2986	3187.0
g15	800	4661	2336.5	2876	2924	194	2907.0	24	2968	3169.0
g16	800	4672	2342.2	2896	2911	190	2914.0	25	2987	3172.0
g17	800	4667	2332.9	2887	2907	194	2909.0	25	2976	-
g18	800	4694	29.7	807	749	242	823.0	63	888	-
g19	800	4661	-59.3	743	634	242	736.0	65	796	-
g20	800	4672	-28.5	728	685	241	768.0	60	829	-
g21	800	4667	-30.8	701	685	241	767.0	65	828	-
g22	2000	19990	9992.1	12402	12699	672	12801.0	120	12956	14123.0
g23	2000	19990	9995.5	12300	12802	660	12783.0	122	12976	14129.0
g24	2000	19990	10012.6	12361	12777	667	12789.0	124	13025	14131.0
g25	2000	19990	9983.3	12393	12616	668	12794.0	122	13015	-
g26	2000	19990	10001.6	12327	12628	674	12787.0	121	12989	-
g27	2000	19990	-42.0	2322	2377	681	2794.0	203	2888	-
g28	2000	19990	-67.3	2294	2441	680	2766.0	193	2877	-

Name	n	m	Random avg	Greedy	Semi-Greedy	Local iters	Local avg	GRASP iters	GRASP best	Known best
g29	2000	19990	36.4	2369	2432	670	2845.0	196	3015	-
g30	2000	19990	54.5	2412	2541	673	2868.0	196	2992	-
g31	2000	19990	-31.3	2334	2438	673	2768.0	202	2881	-
g32	2000	4000	11.7	1026	1002	410	1050.0	34	1156	1560.0
g33	2000	4000	-6.8	1008	1030	409	1024.0	38	1154	1537.0
g34	2000	4000	-22.3	980	1084	409	1016.0	35	1134	1541.0
g35	2000	11778	5903.0	7284	7360	481	7336.0	61	7478	8000.0
g36	2000	11766	5905.5	7222	7359	481	7330.0	59	7463	7996.0
g37	2000	11785	5878.7	7254	7383	475	7334.0	60	7482	8009.0
g38	2000	11779	5898.1	7231	7350	472	7328.0	59	7476	-
g39	2000	11778	7.0	1938	1743	609	1998.0	176	2093	-
g40	2000	11766	-57.5	1896	1705	611	1972.0	159	2070	-
g41	2000	11785	8.2	1876	1692	597	1977.0	158	2089	-
g42	2000	11779	75.7	1967	1649	599	2048.0	162	2146	-
g43	1000	9990	5009.9	6172	6295	334	6393.0	61	6497	7027.0
g44	1000	9990	5001.9	6168	6265	334	6383.0	66	6495	7022.0
g45	1000	9990	4990.1	6200	6315	333	6388.0	63	6516	7020.0
g46	1000	9990	4987.1	6186	6314	330	6380.0	61	6495	-
g47	1000	9990	4987.1	6231	6281	337	6389.0	67	6528	-
g48	3000	6000	3011.5	6000	6000	767	4959.0	1	6000	6000.0
g49	3000	6000	2992.9	6000	6000	761	4943.0	1	6000	6000.0
g50	3000	6000	3007.1	5878	5842	764	4956.0	1	5870	5988.0
g51	1000	5909	2953.2	3623	3686	242	3675.0	32	3753	-
g52	1000	5916	2948.7	3632	3674	239	3680.0	31	3750	-
g53	1000	5914	2951.5	3651	3689	242	3671.0	30	3752	-
g54	1000	5916	2952.8	3633	3682	235	3671.0	29	3748	-